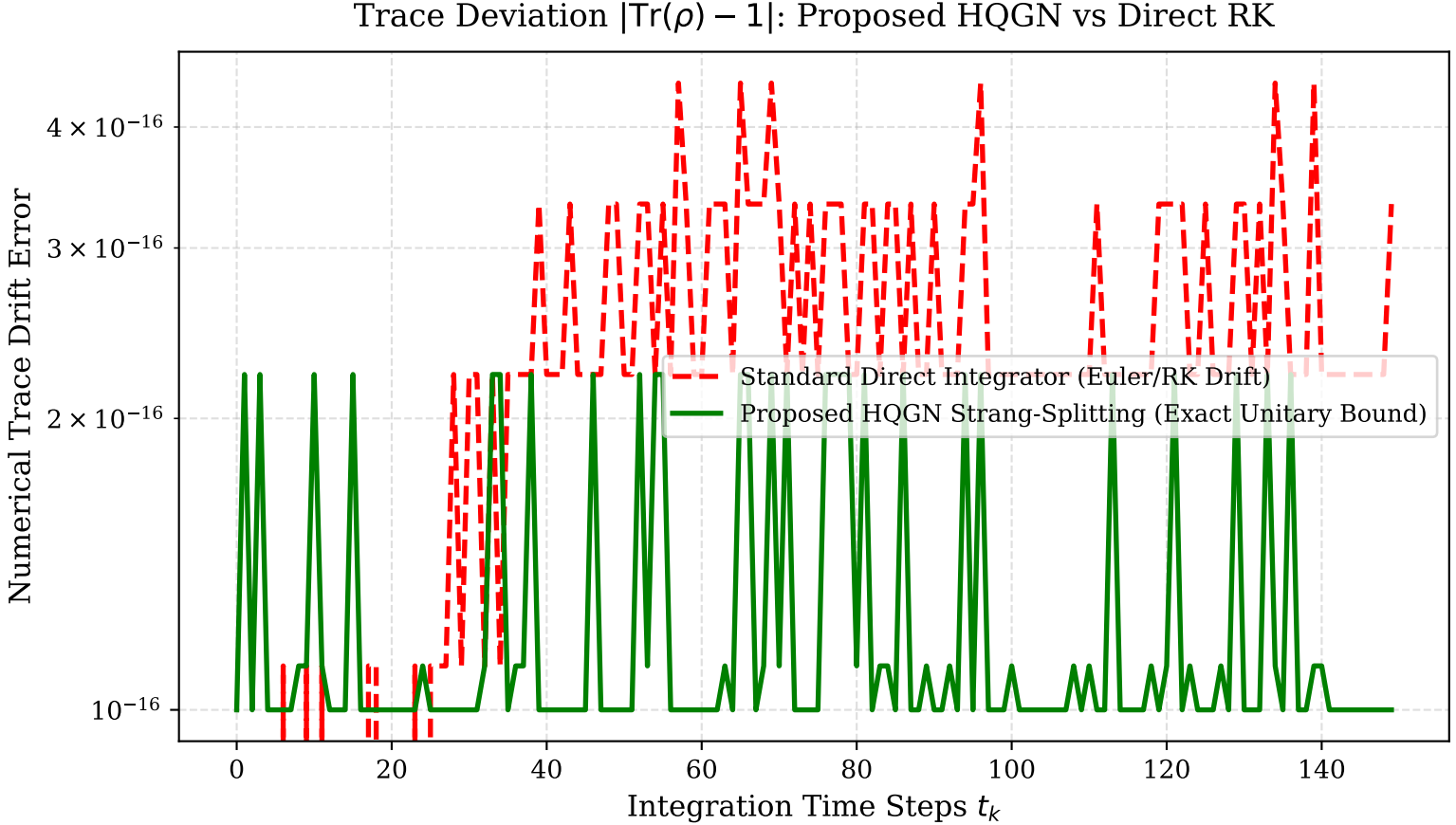
Trace Deviation $|\mathrm{Tr}(\rho) - 1|$: Proposed HQGN vs Direct RK
Numerical Trace Drift Error
4×10^{-16}
3×10^{-16}
2×10^{-16}
10^{-16}
0
20
40
60
80
100
120
140
Integration Time Steps t_k
Standard Direct Integrator (Euler/RK Drift)
Proposed HQGN Strang-Splitting (Exact Unitary Bound)